

NUMBER PATTERNS NOTES

Arithmetic number patterns (linear):

In an arithmetic sequence the **first difference** is constant.

$$\text{Formula: } T_n = a + (n - 1)d$$

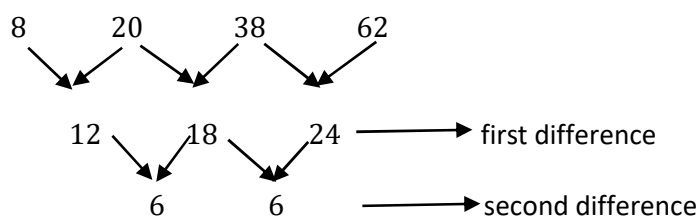
a → the first term of a sequence.

d → the difference between two successive terms.

Quadratic sequences:

In a quadratic sequence the **second difference** is constant.

The first differences of a quadratic sequence form a linear sequence.



$$\text{Formula: } T_n = an^2 + bn + c$$

a → half the second difference ($2a = 2^{\text{nd}}$ difference)

b → a constant

c → a constant

n → position of the n^{th} term in the sequence

NB: You need at least 4 terms before you can assume that you have a quadratic sequence. If you are given 3 terms, the sequence should be seen as a geometric sequence.

Steps to writing an expression for the n^{th} term of a quadratic sequence:

- 1) Calculate the first and second differences. (If the first difference is constant the sequence is linear)
- 2) Calculate a : $2a = \text{second difference}$
- 3) Calculate b : $3a + b = \text{first term of first difference}$
- 4) Calculate c : $a + b + c = \text{first term of the sequence}$

Geometric sequences:

In a geometric sequence the ratio is constant.

$$\text{Formula: } T_n = a \cdot r^{n-1} \quad \text{and} \quad r = \frac{T_2}{T_1} = \frac{T_3}{T_2}$$

a → is the first term

r → ratio

n → position of the n^{th} term in the sequence